# Crash Course in Maths for Cognitive Science

## Session 5 | Probability & Statistics for Modelling

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September 2026

# Session 5 | Probability & Statistics for Modelling

# What today is about

A brisk refresher of the probability and statistics you already met in your bachelor, the foundation the later courses will lean on. Nothing here is new theory; the aim is to reactivate it and line the pieces up.

- *Random variables* and the handful of *distributions* you keep meeting.
- *Expectation* and *variance*, centre and spread.
- The *Normal* distribution, the *law of large numbers*, and the *central limit theorem*.
- *Estimation* and *confidence intervals*; *hypothesis testing*; and *Bayes' theorem*.

# Random variables

**Definition**

A **random variable** $X$ assigns a number to each outcome of a random experiment. It is *discrete* if it takes countably many values (a die, a count of successes) and *continuous* if it takes values in a range (a height, a reaction time).

**Intuition**

The random variable is the "measurement"; the *distribution* is the rule that says how probable each value is. Everything else today is a property of that distribution.

# How a distribution is described

## Useful fact

- **Discrete:** a *probability mass function* $p(x) = \mathbb{P}(X = x)$, with $\sum_x p(x) = 1$.
- **Continuous:** a *probability density* $f(x) \geq 0$ with $\int f(x)\,dx = 1$; probabilities are *areas*, $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx$.
- **Either:** the *cumulative distribution* $F(x) = \mathbb{P}(X \leq x)$, increasing from 0 to 1.

## Common pitfall

For a continuous variable, $\mathbb{P}(X = x) = 0$ for any single point; only intervals carry probability. A density $f(x)$ can even exceed 1; it is an area, not a probability.

# The distributions worth keeping at your fingertips

| **Distribution** | **Models** | **Mean / Variance** |
|---|---|---|
| Bernoulli($p$) | one yes/no trial | $p$ / $p(1-p)$ |
| Binomial($n, p$) | # successes in $n$ trials | $np$ / $np(1-p)$ |
| Poisson($\lambda$) | # rare events in a window | $\lambda$ / $\lambda$ |
| Uniform($a, b$) | "no value preferred" | $\frac{a+b}{2}$ / $\frac{(b-a)^2}{12}$ |
| Normal($\mu, \sigma^2$) | sums of many small effects | $\mu$ / $\sigma^2$ |
| Exponential($\lambda$) | waiting time to an event | $\frac{1}{\lambda}$ / $\frac{1}{\lambda^2}$ |

The first three are discrete, the last three continuous. If you recognise which one a situation calls for, most problems become lookups.

# Binomial and Poisson: a quick reactivation

## Binomial

8 independent trials, each a success with probability $p = 0.25$. The number of successes $X \sim \text{Binomial}(8, 0.25)$ has $\mathbb{P}(X = k) = \binom{8}{k} 0.25^k \, 0.75^{8-k}$, mean $np = 2$, variance $np(1-p) = 1.5$.

## Intuition

When $n$ is large and $p$ small (rare events), the Binomial is well approximated by a **Poisson** with $\lambda = np$, the distribution of counts like spikes per second or arrivals per minute.

# Practice: distributions

**Practice — try it**

1. You flip a fair coin 10 times. Which distribution describes the number of heads, and what are its mean and variance?
2. A call centre receives on average 3 calls per minute. Which distribution fits the count in a given minute?
3. Is "reaction time in milliseconds" a discrete or a continuous variable?

# Expectation: the long-run average

## Definition

The **expectation** (mean) is the probability-weighted average value:

$$\mathbb{E}[X] = \sum_x x\, p(x) \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int x\, f(x)\, dx \quad \text{(continuous)}.$$

## Intuition

$\mathbb{E}[X]$ is where the distribution "balances", the average you would approach if you repeated the experiment many times. It need not be a value $X$ can actually take (the mean of a die is 3.5).

# Variance: the spread

## Definition

The **variance** is the average squared distance from the mean, and the **standard deviation** is its square root (back in the original units):

$$\mathsf{Var}(X) = \mathbb{E}\big[(X-\mu)^2\big] = \mathbb{E}[X^2] - \mu^2, \qquad \mathsf{SD}(X) = \sqrt{\mathsf{Var}(X)}.$$

## Intuition

Variance measures how widely values scatter around the mean. Two distributions can share a mean yet look completely different because their variances differ.

# The rules you actually use

**Useful fact**

- **Linearity of expectation** (always, even for dependent variables): $\mathbb{E}[aX + bY + c] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y] + c$.
- **Scaling/shift of variance:** $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$, a shift does nothing, a scale squares.
- **Sums of *independent* variables:** $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

## Expectation and variance: try it

**Example**

Let $X$ be the result of one roll of a fair six-sided die. Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

*Try it yourself first — the solution is on the next slide.*

# Expectation and variance: solution

Each face has probability $\frac{1}{6}$, so

$$\mathbb{E}[X] = \tfrac{1}{6}(1+2+3+4+5+6) = \tfrac{21}{6} = 3.5.$$

For the variance, $\mathbb{E}[X^2] = \frac{1}{6}(1+4+9+16+25+36) = \frac{91}{6}$, hence

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu^2 = \frac{91}{6} - 3.5^2 = \frac{91}{6} - \frac{147}{12} = \frac{35}{12} \approx 2.92.$$

So $\mathrm{SD}(X) \approx 1.71$, the typical distance of a roll from the average 3.5.

# Practice: expectation and variance

**Practice — try it**

1. For $X \sim \text{Bernoulli}(p)$, show $\mathbb{E}[X] = p$ and $\text{Var}(X) = p(1-p)$.
2. If $\text{Var}(X) = 4$, what is $\text{Var}(3X+5)$?
3. Two independent variables have variances 2 and 3. What is the variance of their sum?

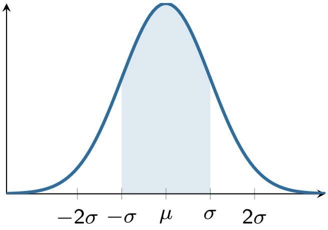
# The Normal distribution

**Definition**

$X \sim \mathcal{N}(\mu, \sigma^2)$ has the bell-shaped density

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

$\mu$ sets the centre, $\sigma$ the width. Its famous rule: about 68% / 95% / 99.7% of the mass lies within 1 / 2 / 3 standard deviations of the mean.




The central 68% band ($\pm 1\sigma$).

# Standardising: the $z$-score

**Key idea**

Subtract the mean and divide by the standard deviation to get a **standard normal** $Z \sim \mathcal{N}(0,1)$:

$$Z = \frac{X - \mu}{\sigma}.$$

A $z$-score says how many standard deviations a value sits from the mean, so a single table of $Z$ answers questions about *any* Normal.

# The $z$-score: try it

**Example**

IQ scores are modelled as $\mathcal{N}(100, 15^2)$. What fraction of people score above 130?

*Try it yourself first — the solution is on the next slide.*

# The $z$-score: solution

Standardise the threshold:

$$z = \frac{130 - 100}{15} = 2.$$

So we need $\mathbb{P}(Z > 2)$. From the 68–95–99.7 rule, 95% lies within $\pm 2\sigma$, leaving 5% in the two tails, or about 2.5% in each. The exact value is $\mathbb{P}(Z > 2) = 2.28\%$, roughly one person in forty.

# Two theorems that make statistics work

## Law of large numbers

As the sample size grows, the **sample mean** $\bar{X}_n$ converges to the true mean $\mu$. Averaging many observations cancels the noise.

## Central limit theorem

For large $n$, the sample mean is approximately **Normal**,

$$\bar{X}_n \;\approx\; \mathcal{N}\!\left(\mu,\; \frac{\sigma^2}{n}\right),$$

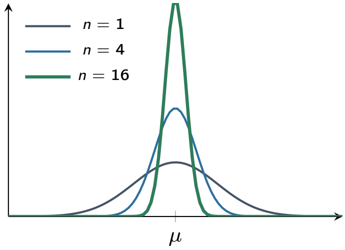
*whatever* the shape of the original distribution. Its standard deviation, the **standard error**, is $\mathrm{SE} = \sigma/\sqrt{n}$.

# Why the standard error shrinks like $1/\sqrt{n}$

**Intuition**

The LLN says the average settles down; the CLT says *how fast* and *in what shape*. Because $\text{SE} = \sigma/\sqrt{n}$, quadrupling the sample only halves the error, precision is expensive, but guaranteed.

This is the engine behind every confidence interval and test that follows: the sampling distribution of the mean is (approximately) Normal with a known width.




The mean's distribution narrows as $n$ grows.

# Practice: Normal, LLN, CLT

## Practice — try it

1. A score is 1.5 standard deviations above the mean. What is its $z$-score, and is it in the top 10%?
2. Measurements have $\sigma = 15$. What is the standard error of the mean of $n = 25$ of them?
3. You average a very skewed variable over a large sample. What shape does the average have, approximately?

# From a sample to an estimate

## Definition

An **estimator** is a rule that turns a sample into a guess for an unknown quantity. The two everyday ones are the **sample mean** and **sample variance**:

$$\bar{x} = \frac{1}{n}\sum_i x_i, \qquad s^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2.$$

## Intuition

An estimate is itself random; a different sample gives a different number. So we care not just about the estimate but about how much it would wobble from sample to sample.

# Bias, and the mysterious $n - 1$

**Key idea**

An estimator is **unbiased** if it is right *on average* over many samples. The sample mean is unbiased for $\mu$. Dividing the squared deviations by $n$ would *underestimate* the variance, because deviations are taken about the sample mean (which sits inside the data). Dividing by $n - 1$ corrects exactly for this and makes $s^2$ unbiased.

The "$-1$" is the one *degree of freedom* spent estimating the mean.

# Confidence intervals

**Definition**

A 95% **confidence interval** is a range built from the data so that, over repeated samples, 95% of such intervals contain the true parameter. For a mean with known $\sigma$ it is

$$\bar{x} \;\pm\; 1.96\,\frac{\sigma}{\sqrt{n}} \qquad \text{(1.96 from the Normal).}$$

**Common pitfall**

The interval is random, not the parameter. "95% confidence" describes the *procedure* (95% of intervals catch the truth), not a probability that this one particular interval does.

# A confidence interval: try it

**Example**

From $n = 25$ measurements with known $\sigma = 15$ you obtain a sample mean $\bar{x} = 100$. Give a 95% confidence interval for the true mean $\mu$.

*Try it yourself first — the solution is on the next slide.*

# A confidence interval: solution

The standard error is $\mathrm{SE} = \sigma/\sqrt{n} = 15/5 = 3$, so the interval is

$$100 \ \pm \ 1.96 \times 3 \ = \ 100 \ \pm \ 5.88 \ = \ [94.1, \ 105.9].$$

**Intuition**

Wider $\sigma$ or smaller $n$ widens the interval; more data $(n \uparrow)$ shrinks it like $1/\sqrt{n}$. If $\sigma$ is unknown we estimate it by $s$ and use the $t$-distribution instead of 1.96, slightly wider, to pay for the extra uncertainty.

# The logic of a test

## Key idea

State a **null hypothesis** $H_0$ ("nothing is going on") and an **alternative** $H_1$. *Assume* $H_0$, then ask: how surprising is the data we actually saw? If it is too surprising under $H_0$, we reject $H_0$.

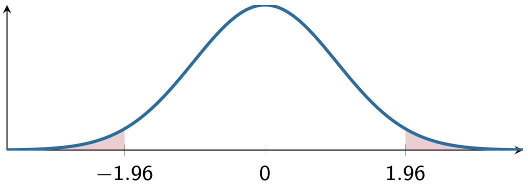
## Intuition

A hypothesis test is a proof by contradiction with a probability attached. We never prove $H_0$ true; we only find the data compatible with it, or surprising enough to doubt it.

# Test statistic, $p$-value, significance

## Definition

A **test statistic** measures how far the data sit from what $H_0$ predicts, in standard-error units. The $p$**-value** is the probability, *if $H_0$ were true*, of a statistic at least this extreme. Reject $H_0$ when $p < \alpha$, the pre-chosen **significance level** (often 0.05).




Two-tailed test at $\alpha = 0.05$: reject if the statistic lands in a red tail ($|z| > 1.96$).

# A one-sample test: try it

**Example**

A standard task has mean 100. A group of $n = 25$ people scores $\bar{x} = 106$; the population standard deviation is $\sigma = 15$. Test $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$ at $\alpha = 0.05$.

*Try it yourself first — the solution is on the next slide.*

# A one-sample test: solution

The standard error is $\mathrm{SE} = 15/\sqrt{25} = 3$, so the test statistic is

$$z = \frac{\bar{x} - \mu_0}{\mathrm{SE}} = \frac{106 - 100}{3} = 2.0.$$

Since $|z| = 2.0 > 1.96$, we reject $H_0$ at $\alpha = 0.05$ (two-tailed $p \approx 0.046$).

**Intuition**

The score is two standard errors above the null value, surprising enough, if nothing were going on, to doubt $H_0$. Note the group could still differ by a trivial amount: *statistical* significance is not *practical* importance.

# Two ways to be wrong

### Useful fact

- **Type I error** ($\alpha$): reject $H_0$ when it is actually true, a false alarm.
- **Type II error** ($\beta$): fail to reject $H_0$ when $H_1$ is true, a miss.
- **Power** $= 1 - \beta$: the chance of catching a real effect. It grows with sample size and effect size.

| | $H_0$ true | $H_1$ true |
|---|---|---|
| reject $H_0$ | Type I | ✓ power |
| keep $H_0$ | ✓ | Type II |

### Common pitfall

A large $p$-value is *not* evidence that $H_0$ is true, only a failure to reject it. And a small $p$ tells you an effect is unlikely due to chance, not that it is large or important.

# Practice: estimation and testing

## Practice — try it

1. You double the sample size. By what factor does a confidence interval's width change?
2. A test gives $p = 0.20$. Have you shown there is no effect?
3. Name one way to increase the power of a test.

# Conditional probability, and reversing it

## Definition

The **conditional probability** of $A$ given $B$ is $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$. Rearranged, $\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$.

## Intuition

Often we know one direction, $\mathbb{P}(\text{positive test} \mid \text{disease})$, but want the other, $\mathbb{P}(\text{disease} \mid \text{positive test})$. Bayes' theorem is the tool that flips the conditioning around.

# Bayes' theorem

**Key idea**

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid \bar{A})\mathbb{P}(\bar{A}).$$

The denominator (the *total probability* of $B$) just adds up all the ways $B$ can happen. The numerator keeps the one path through $A$. Their ratio is the updated probability of $A$ once we have learned $B$.

# The base-rate example: try it

**Example**

A disease affects 1% of people. A test detects it in 99% of those who have it (sensitivity), but also gives a false positive in 5% of those who do not. You test positive. What is the probability you actually have the disease?

*Try it yourself first — the solution is on the next slide.*

# The base-rate example: solution

Total probability of a positive test:

$$\mathbb{P}(+) = \underbrace{0.99 \cdot 0.01}_{\text{true } +} + \underbrace{0.05 \cdot 0.99}_{\text{false } +} = 0.0594.$$

Bayes' theorem then gives

$$\mathbb{P}(D \mid +) = \frac{0.99 \cdot 0.01}{0.0594} \approx 0.17.$$

Only about **17%**, because the disease is rare, most positives are false positives. This *base-rate* effect is the classic surprise of Bayes.

| per 10,000 | test + | test − |
|---|---|---|
| disease (100) | 99 | 1 |
| healthy (9900) | 495 | 9405 |
| total | 594 | 9406 |

Of 594 positives, only 99 are truly ill: $99/594 \approx 17\%$.

# Prior and posterior: the idea in one line

**Intuition**

Bayes' theorem turns a **prior** probability (the 1% base rate, before testing) into a **posterior** probability (the 17%, after seeing the positive). New evidence updates belief, and the base rate you started from matters just as much as the test result.

That single move, prior, times evidence, renormalised, is the whole of Bayesian reasoning, and you will meet it again throughout data analysis.

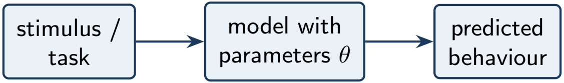
# Practice: Bayes

**Practice — try it**

1. Write Bayes' theorem for $\mathbb{P}(A \mid B)$ from memory.
2. In the disease example, if the base rate rose to 10%, would $\mathbb{P}(D \mid +)$ go up or down?
3. Why can a very accurate test still give many false positives for a rare condition?

# A glimpse ahead: modelling cognition

Everything so far *describes* data. A **cognitive model** goes one step further: it proposes *how* the data were generated, a mechanism with a few **parameters** we can estimate from behaviour.




**Intuition**

The parameters *are* the psychology: a learning rate, a decision threshold, a perceptual bias, a memory-decay constant. Fitting the model estimates them from data; comparing models asks which mechanism explains behaviour best.

# The modelling pipeline

The tools you just refreshed slot into one loop, the workflow of the modelling courses ahead:

data → model $p(x \mid \theta)$ → fit $\theta$ → uncertainty → compare & predict

**Intuition**

Distributions describe the data; the *likelihood* fits the parameters; the *central limit theorem* and *confidence intervals* quantify uncertainty; and *Bayes* lets prior knowledge join in. You already own every piece.

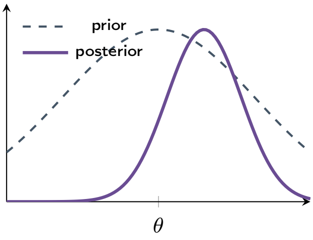
# Bayes, one level up: inference over a parameter

The Bayes rule you just used for *events* works identically for a *parameter* $\theta$:

$$\underbrace{p(\theta \mid D)}_{\text{posterior}} \propto \underbrace{p(D \mid \theta)}_{\text{likelihood}} \times \underbrace{p(\theta)}_{\text{prior}}.$$

**Intuition**

Now the answer is a whole *distribution* over $\theta$: our belief after seeing the data. A flat prior and lots of data give a sharp posterior; little data leaves it broad, uncertainty made visible.




Data sharpen a belief about $\theta$.

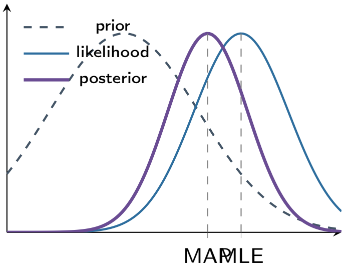
# Two point estimates: MLE and MAP

If you want a single number from all this:

- **MLE**, the $\theta$ at the peak of the *likelihood* (best fit to the data alone).
- **MAP**, the $\theta$ at the peak of the *posterior* (likelihood $\times$ prior).

**Intuition**

MAP is just the MLE *nudged by the prior*. With a flat prior they coincide; with little data the prior pulls the estimate toward what is reasonable; with lots of data the likelihood dominates and MAP $\to$ MLE.




Prior $\times$ likelihood $=$ posterior; MAP sits between prior and MLE.

## Session 5: what we refreshed

- *Random variables* and the core *distributions*; *expectation* and *variance* as centre and spread.
- The *Normal* distribution and $z$-scores; the *law of large numbers* and the *central limit theorem*, with $\mathrm{SE} = \sigma/\sqrt{n}$.
- *Estimation* and *confidence intervals*; *hypothesis testing* with $p$-values and the two error types.
- *Bayes' theorem*, and the base-rate lesson that priors matter.
- a *glimpse ahead*: the modelling pipeline, Bayes over parameters, and MLE vs MAP.

These are the working tools you will reach for across data analysis, statistics, and the modelling courses ahead.